

Programming FTP Clients with Python

This article shows how readers can write their own FTP client programs, using Python.



There's a lot happening on the Internet these days, and transferring files is a very popular activity that takes place all the time. One popular way to accomplish this is by using the FTP (File Transfer Protocol). This is a standard Internet protocol for transmitting files between computers on the Internet. There are several other protocols used over the Web like HTTP, SMTP and scp/rsync, but they serve different purposes. Hypertext Transfer Protocol (HTTP) transfers displayable Web pages and related files, and the Simple Mail Transfer Protocol (SMTP) transfers e-mail. FTP is an application protocol that uses the Internet's TCP/IP protocols. It is built on client-server architecture. FTP is most commonly used to download a file from a server using the Internet or to upload a file to a server (e.g., uploading a Web page file to a server). FTP users may authenticate themselves using a clear-text sign-in protocol, normally in the form of a user name and password, but can connect anonymously if the server is configured to allow it. So while FTP supports both anonymous logins as well as authenticated logins, even with anonymous login support, it is restrictive and secure.

How it works

In this article, we will look at how to write our own FTP client programs. Before that, let's look at how the process works.

The protocol is shown as a diagram in Figure 1 and works as follows:

1. The client contacts the FTP server on the remote host.
2. The client logs in with a user name and password (or anonymously with an e-mail address).
3. The client performs various file transfers or information requests.

4. The client completes the transaction by logging out of the remote host and FTP server.

Python and FTP

Normal file transfers are carried out through the command line or with the use of a graphical interface program. A Web browser can also make FTP requests to download programs you select from a Web page. But as programmers, we may need to write our own FTP client side programs.

Python supports most Internet protocols, including FTP. Other supported client libraries can be found at <http://docs.python.org/library/internet>.

ftplib, the FTP protocol client, is available with Python. This module defines the FTP class and a few related items. The FTP class implements the client side of the FTP protocol. You can use this to write Python programs that perform a variety of automated FTP jobs, such as mirroring other FTP servers.

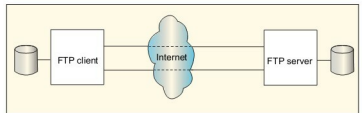


Figure 1: FTP flow diagram

Let's take a look at the simple steps to create an Internet client with Python. This task is made pretty simple by the Python *ftplib* module. By importing this module and making appropriate calls, our job is pretty much done. The steps